Q9. Program to check for leap year?

```
# Taking input from user
year <- as.integer(readline(prompt = "Enter a year: "))
# Checking leap year
if ((year %% 4 == 0 && year %% 100 != 0) || (year %% 400 == 0)) {
  cat(year, "is a Leap Year")
} else {
  cat(year, "is not a Leap Year")
}
```

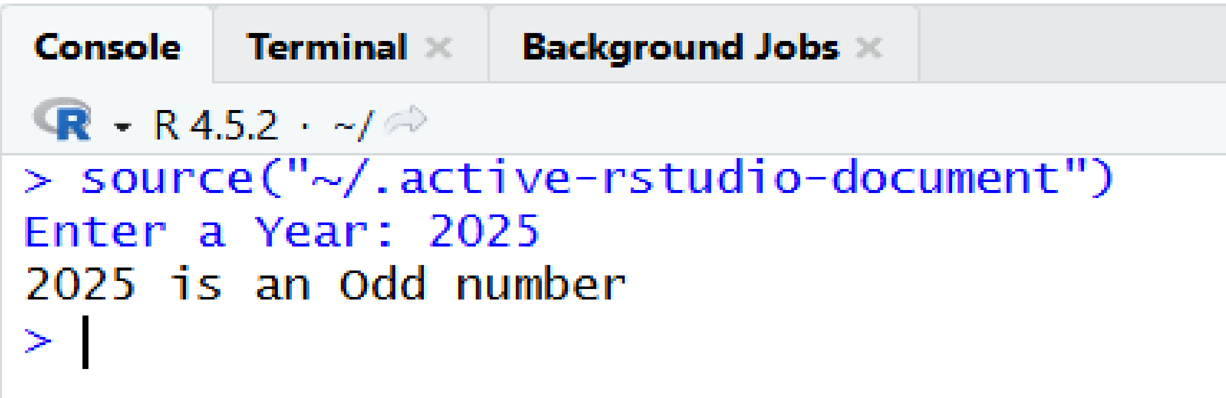
Output:

```
Console   Terminal   Background Jobs
R 4.5.2 · ~/
> source("~/.active-rstudio-document")
Enter a year: 2024
2024 is a Leap Year
>
```

Q10. Check whether number is odd or even?

```
# Taking input from user
num <- as.integer(readline(prompt = "Enter a Year: "))
# Checking odd or even
if (num %% 2 == 0) {
  cat(num, "is an Even number")
} else {
  cat(num, "is an Odd number")
}
```

Output:



Q11. Check if a number is positive negative or zero?

```
# Taking input from user
num <- as.numeric(readline(prompt = "Enter a number: "))
# Checking the number
if (num > 0) {
  cat(num, "is a Positive number")
} else if (num < 0) {
  cat(num, "is a Negative number")
} else {
  cat("The number is Zero")
}
```

Output:

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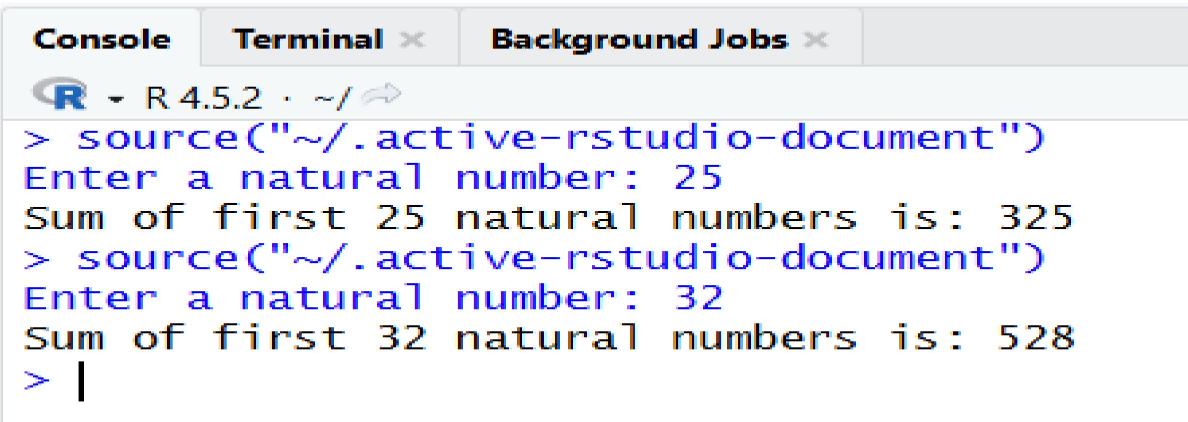
```
> source("~/.active-rstudio-document")
Enter a number: 50
50 is a Positive number
> source("~/.active-rstudio-document")
Enter a number: -90
-90 is a Negative number
> source("~/.active-rstudio-document")
Enter a number: 0
The number is Zero
>
```

Q12. Find the sum of Natural Numbers?

```
# Taking input from user

n <- as.integer(readline(prompt = "Enter a natural number: "))

# Finding sum of first n natural numbers

sum <- 0

for (i in 1:n) {

  sum <- sum + i

}

# Displaying result

cat("Sum of first", n, "natural numbers is:", sum)
```

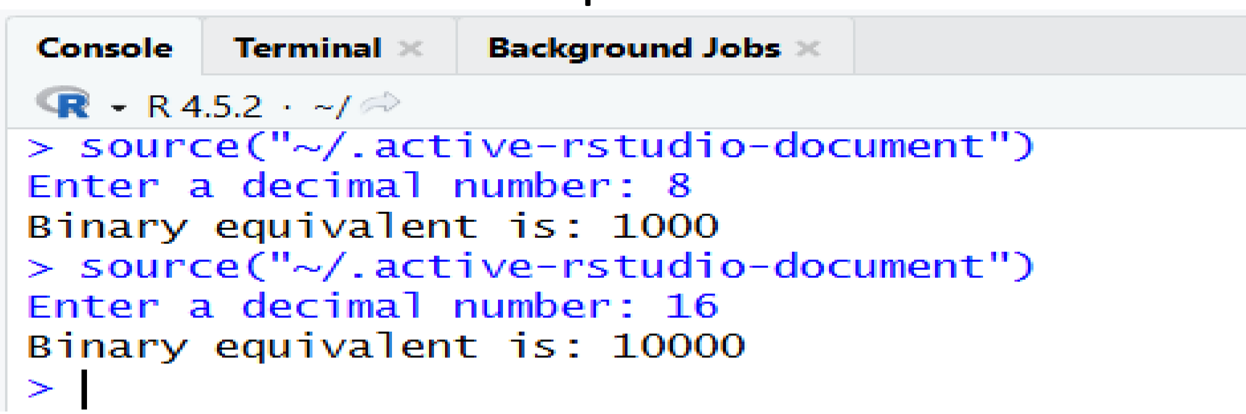
Output:



Q13. Convert Decimal into Binary using Recursion?

```
# Recursive function to convert decimal to binary
decimalToBinary <- function(n) {
  if (n == 0) {
    return("")
  } else {
    return(paste0(decimalToBinary(n %/% 2), n %% 2))
  }
}
# Taking input from user
num <- as.integer(readline(prompt = "Enter a decimal number: "))
if (num == 0) {
  cat("Binary equivalent is: 0")
} else {
  binary <- decimalToBinary(num)
  cat("Binary equivalent is:", binary)
}
```

Output:



Q14. Find the factorial of number using Recursion?

```
# Recursive function to find factorial
factorial_rec <- function(n) {
  if (n == 0 || n == 1) {
    return(1)
  } else {
    return(n * factorial_rec(n - 1))
  }
}
# Taking input from user
num <- as.integer(readline(prompt = "Enter a number: "))
if (num < 0) {
  cat("Factorial is not defined for negative numbers")
} else {
  result <- factorial_rec(num)
  cat("Factorial of", num, "is:", result)
}
```

Output:

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> source("~/.active-rstudio-document")
Enter a number: 5
Factorial of 5 is: 120
> source("~/.active-rstudio-document")
Enter a number: 12
Factorial of 12 is: 479001600
>
```

Q15. Program to find HCF or GCD of a number?

```
# Recursive function to find GCD
gcd <- function(a, b) {
  if (b == 0) {
    return(a)
  } else {
    return(gcd(b, a %% b))
  }
}
# Taking input from user
num1 <- as.integer(readline(prompt = "Enter first number: "))
num2 <- as.integer(readline(prompt = "Enter second number: "))
# Finding GCD
result <- gcd(abs(num1), abs(num2))
# Displaying result
cat("GCD of", num1, "and", num2, "is:", result)
```

Output:

```
Console   Terminal   Background Jobs
R 4.5.2 · ~/
> source("~/.active-rstudio-document")
Enter first number: 36
Enter second number: 60
GCD of 36 and 60 is: 12
> 
```